Identification of ZIP Code Areas Eligible for Medicare HPSA Bonus Payments

DOCUMENTATION
October 30, 2014

BACKGROUND

The Medicare Prescription Drug, Improvement, and Modernization Act of 2003 (MMA 2003) amended the Social Security Act provisions for bonus payments for physician services provided in geographic Health Professional Shortage Areas (HPSAs), to make such payments automatic rather than requiring the physician to apply for them. Automatic bonus payments are provided to physicians in Primary Care (PC) and Mental Health (MH) HPSAs with designation types of Single County or Service Area. Medicare HPSA bonus payments apply to all physicians who perform covered services within a primary medical care HPSA. All eligible physicians receive bonus payments regardless of specialty. In addition, psychiatrist furnishing services in mental health HPSAs are eligible to receive incentive payments. Medicare, however, does not pay bonuses for services provided in Dental Care (DC) HPSAs. Centers for Medicare and Medicaid Services (CMS) will use dental HPSAs to determine Electronic Health Record (EHR) incentive payments, per the American Recovery & Reinvestment Act of 2009 (ARRA).

To facilitate identification of eligible claims, the purpose of this contract is to identify ZIP Codes that correspond to areas where physicians qualify for bonus payments. Because ZIP Code geography does not have any direct correspondence with the boundaries of HPSA regions, ZIP Code lists have been produced for automatic bonus payment if the ZIP Code area is either totally within one or more qualifying HPSA region, or, in the case of Single County HPSAs, the ZIP Code has been assigned to that county as dominant according to the United States Postal Service (USPS).

DELIVERABLE FILES

In total, four lists of ZIP Codes were produced with three files as the deliverables that indicate payment status as follows:

1) ***"FullyDentalCareHPSA_20140829.xlsx:"*** Contains ZIP Codes for Dental Care HSPA status. The file lists ZIP Codes and the state each ZIP Code is located in. The listed ZIP Codes were selected based on a two part analysis.
    a. First, all ZIP Codes were geographically mapped with all Dental Care HPSA boundaries, and a percentage of area overlap was calculated and assigned to each ZIP Code. The first eligibility selection criteria was a 99% area overlap or greater.
    b. Second, a list (provided by CMS) of ZIP Codes that the USPS considers dominant to a Single County Dental Care HPSA was used. ZIP Codes on the dominant county list and in counties identified by FIPS Codes that were within designated HPSAs were flagged and selected regardless of the geographic area overlap percentage calculated.

"FullyDentalCareHPSA_20140829.xlsx" has 4,018 records. The file record layout is:

Field	Field_Name	Description/Comments	Data Type
1	ZIP CODE	5-Digit ZIP Code	Char(5)
2	STATE	State Abbreviation	Char(2)

2) ***"FullyMentalHealthHPSA_20140829.xlsx:"*** Contains ZIP Codes that qualify for automatic payment for Mental Health HSPA status. The file lists ZIP Codes and the state each ZIP Code is located in. The listed ZIP Codes were selected based on a two part test.
    a. First, all ZIP Codes were geographically mapped with all Mental Health HPSA boundaries, and a percentage of area overlap was calculated and assigned to each ZIP Code. The first eligibility selection criteria was a 99% area overlap or greater.
    b. Second, a list (provided by CMS) of ZIP Codes that the USPS considers dominant to a Single County Mental Health HPSA was used. ZIP Codes on the dominant county list and in counties identified by FIPS Codes that were within designated HPSAs were flagged and selected regardless of the geographic area overlap percentage calculated.

"FullyMentalHealthHPSA_20140829.xlsx" has 19,058 records. The file record layout is:

Field	Field_Name	Description/Comments	Data Type
1	ZIP CODE	5-Digit ZIP Code	Char(5)
2	STATE	State Abbreviation	Char(2)

3) ***"FullyPriCareHPSA_20140829.xlsx:"*** Contains ZIP Codes that qualify for automatic payment for Primary Care HSPA status. The file lists the ZIP Codes and the state each ZIP Code is located in. The listed ZIP Codes were selected based on a two part analysis.
    a. First, all ZIP Codes were geographically mapped with all Primary Care HPSA boundaries, and a percentage of area overlap was calculated and assigned to each ZIP Code. The first eligibility selection criteria was a 99% area overlap or greater.
    b. Second, a list (provided by CMS) of ZIP Codes that the USPS considers dominant to a Single County Primary Care HPSA was used. ZIP Codes on the dominant county list and in counties identified by FIPS Codes that were within designated HPSAs were flagged and selected regardless of the geographic area overlap percentage calculated.

"FullyPriCareHPSA_20140829.xlsx" has 7,141 records. The file record layout is:

Field	Field_Name	Description/Comments	Data Type
1	ZIP CODE	5-Digit ZIP Code	Char(5)
2	STATE	State Abbreviation	Char(2)

The fourth file, ***"USzips20140829_D_ALL.xlsx,"*** contains detailed ZIP Code and pertinent Census boundary information, the USPS dominant county status, and the calculated percentages of geographic overlap. This information may be needed if questions are received from health care providers or the public. This document is available for CMS review upon request.

"USzips20140829_D_ALL.xlsx" has 41,363 records. The file record layout is:

Field	Field_Name	Description/Comments	Data Type
1	ZIP	5-Digit ZIP Code	Char(5)
2	Enclosing_Zip	5-Digit Enclosing ZIP boundary	Char(5)
3	Name	Last Line Name of Preferred Postal Place Name	Char(100)
4	Multiple_Name	Flag identifying name in Multiple Names table	Logical
5	Name_Lng	Language code	Char(3)
6	Name_Type	LL: Last Line Name of Preferred Postal Place Name	Char(2)
7	Feature_Type	Feature Type	Small Int
8	Area_Meters	Area in meters	Float
9	Perimeter_Meters	Perimeter in meters	Float
10	Latitude	Centroid of Latitude	Float
11	Longitude	Centroid of Longitude	Float
12	State	State Abbreviation	Char(2)
13	State_FIPS	Official State FIPS code	Char(2)
14	County	County Name	Char(100)
15	County_FIPS	County FIPS	Char(5)
16	Block	Census Block FIPS	Char(18)
17	Tract	Census Tract FIPS	Char(14)
18	BlockGroup	Census Block Group FIPS	Char(15)
19	CBSA	Metropolitan and Micropolitan Statistical Area Name	Char(100)
20	CBSA_Code	Metropolitan and Micropolitan Statistical Area Code	Char(5)
21	MCD	Minor Civil Division Name	Char(75)
22	MCD_Code	Minor Civil Division Code	Char(10)
23	Transportation_ID	Transportation Element ID	Decimal(15,0)
24	Side_of_Line	Side of Line Network feature	Small Int
25	Relative_Position	The relative position along the Network feature beginning at the from junction	Decimal(8,2)
26	ID	TomTom	Decimal(15,0)
27	USPS_Dom_SinglCo	Flag for ZIP being USPS Dominant to HPSA County (0/1)	Logical
28	Pct_In_Dental	Percentage of area overlap calculated with all Dental Care HPSAs	Float
29	Pct_or_Dom_Dental	Percentage of area overlap calculated with all Dental Care HPSAs or 100% for ZIP being USPS Dominant to HPSA County	Float
30	Pct_In_Mental	Percentage of area overlap calculated with all Mental Health HPSAs	Float
31	Pct_or_Dom_Mental	Percentage of area overlap calculated with all Mental Health HPSAs or 100% for ZIP	Float

		being USPS Dominant to HPSA County	
32	Pct_In_Primary	Percentage of area overlap calculated with all Primary Care HPSAs	Float
33	Pct_or_Dom_Primary	Percentage of area overlap calculated with all Primary Care HPSAs or 100% for ZIP being USPS Dominant to HPSA County	Float

The above file only includes ZIP Codes for the U.S. (50 states), Puerto Rico, and new in 2014 are Guam and the U.S. Virgin Islands. For U.S. outlying territories, we performed our analysis separately. Below is a table for "Designated" HRSA status with "Single County" counts for outlying territories we extracted from the HRSA designation file (*2014-08-29_All_Geo_HPSAs.xlsx*):

State FIPS	HPSA Primary State	State Abbreviation	Dental Care ("Single County")	Mental Health ("Single County")	Primary Care ("Single County")	Zip Codes to Append
60	American Samoa	AS	5	5	5	1 DC, 1 MH, 1 PC
64	Federated States of Micronesia	FM	4	4	3	4 DC, 4 MH, 4 PC
66	Guam	GU	0	0	1	13 PC
68	Marshall Islands	MH	0	0	31	2 PC
69	Northern Mariana Islands	MP	4	4	4	3 PC, 3 DC, 3 MH
70	Republic of Palau	PW	16	16	16	2 PC, 2 DC, 2 MH
78	U.S. Virgin Islands	VI	3	3	3	16 DC, 16 MH, 16 PC

The total Zip Code counts added for outlying regions are 26 Dental Care; 26 Mental Health; 41 Primary Care.

STEPS FOR CREATING THE DELIVERABLE FILES

To create the deliverable files, the following tasks were completed.

1) Review HPSA Definition File

The HPSA data was downloaded from http://datawarehouse.hrsa.gov/data/datadownload/hpsadownload.aspx. Data and documentation was approx. 2.3 MB. A single file, HRSA.zip, contained the data for Primary Care, Dental Care, and Mental Health HPSAs as separate text files, and also included documentation. The latest available text files dated August 29, 2014 were used.

The file includes the field **HPSA_Status_Description** indicating which claims are currently eligible for bonus payments. The **HPSA_Status_Description** types provided are “Designated” and “Proposed Withdrawal.” The ZIP Codes delivered from our analysis only use the “Designated” or “Proposed Withdrawal” types of **HPSA_Status_Description**, not “Withdrawn.”

The file includes three **Discipline_Class_Description** types for “Primary Care,” “Mental Health,” and “Dental Care.”

The file contains one record for each geographic component defining a HPSA area in field **HPSA_Component_Type_Description**. Geographic units that can compose a HPSA include counties, census tracts, minor civil divisions (MCDs, also known as county subdivisions or townships), and Unknown. The “Unknown” records were used. They all matched State+County FIPS codes so they were treated as counties. Designations can be composed of any of these geographic units, alone or in combination (e.g., a county and an MCD). The HRSA Data Warehouse HPSA records contain information about both the geographic component of the HPSA and the whole HPSA designation.

The file includes **HPSA_Type_Description** types for Geographical Area, Single County, Population Group, and several others. The ZIP Codes delivered from our analysis only use the “Geographical Area” and “Single County” types of **HPSA_Status_Description**

The Excel file was verified against the data files downloaded from HRSA data warehouse using the SAS system statistical software. The data files were cleaned and joined using the combination of discipline type and service_area_code. Unneeded records (for the dental discipline, types including population groups and facilities, and HPSAs not currently designated) were eliminated. Comparison results were saved into a separate file for analysis.

2) Acquire Boundary Files

Mapping software, or geographic information systems need geographic boundary files to represent the geographic extent of the real-world areas being modeled as polygons. We used the Geographic Information System (GIS) **MapInfo Professional** for this analysis.

Boundaries/polygons for counties, census tracts, and MCDs were needed because these are the geographic units that are used to define Single County and Service Area/Geographic HPSAs. The definition file *“HPSA.zip”* used 2010 Census Geographic ID codes representing Single Counties, Census Tracts, and MCDs. The HRSA Data Warehouse uses 2010 Census boundary files for the vast majority of boundaries with only a few2000 Census boundaries. 1990 Census boundaries no longer represent any Designated HRSA boundaries.

The following geographic files were downloaded from www.census.gov:

2010 County and County Equivalent Areas
Downloadable as 3 individual zip files:
All 50 States, D.C., and Puerto Rico, tl_2010_us_county10.zip (72,703 kb)
Guam, tl_2010_66_county10.zip (14,799 kb)
Virgin Islands, tl_2010_78_county10.zip (13,890 kb)
http://www.census.gov/cgi-bin/geo/shapefiles2010/main
Census metadata lists Projection = Geographic (Lat/Lon), Datum = NAD83

2010 County Subdivisions and 2010 Sub Barrios (Puerto Rico Only)
Downloadable as 56 individual zip files:
Alabama - tl_2010_01_cousub10.zip (7,321 kb)
Through
Virgin Islands - tl_2010_78_cousub10.zip (2,852 kb)
http://www.census.gov/cgi-bin/geo/shapefiles2010/main
Census metadata says Projection = Geographic (Lat/Lon), Datum = NAD83

Census 2000: Census Tract
Downloadable as 57 individual zip files
Alabama - tl_2010_01_tract10.zip (9,862 kb)
Through
Virgin Islands - tl_2010_78_tract10.zip (378 kb)
http://www.census.gov/geo/www/cob/tr2010.html
Census metadata says Projection = Geographic (Lat/Lon), Datum = NAD83

5-Digit ZIP Codes
The 5-Digit ZIP Codes product from Pitney Bowes Business Intelligence (PBBI/MapInfo) is produced by TomTom (formerly TeleAtlas). The ZIP Code data covers all 50 United States, Puerto Rico, the District of Columbia, Guam, and the U.S. Virgin Islands. The 5-Digit ZIP Code product constitutes a geographic data set that contains the boundaries for each 5-Digit ZIP Code in the United States assigned by the U.S. Postal Service, but without blank space for water features such as lakes and rivers. MCD (Minor Civil Division) and CBSA (Core-Based Statistical Area) codes are included, in addition to State, Census Tract, Block Group and Block information. All coordinates are based on WGS (World Geodetic System) 84. All longitude and latitude coordinates are accurate to six decimal places of precision.

PBBI's Postal Products are updated quarterly. The product media explicitly shows that the 'vintage' of the source data used for this project was June 2014 (2014.06). 'Vintage' refers to the currency of the data as being of a specific date.

3) Prepare HPSA_Designations Excel File

The three text files (HPSA_DC.txt, HPSA_MH.txt, and HPSA_PC.txt) contained in HRSA.zip were converted to Excel files per the instructions and then saved as MapInfo files for processing. Record counts were noted so that the total count could be maintained throughout the disaggregation and reassembly process. Fields were added to facilitate splitting the data and putting the tables back together. A **GIS_Boundary_Source** field was added to identify which HPSA elements were matched with the Census 2010 geographies. Minor formatting issues were corrected that caused a few records to have carriage returns before the end of the records. The three text files had 81,524 records. The **HPSA_Status_Description** field had 39,075 "Withdrawn" records, 18 "Proposed Withdrawal" records, and 42,431 "Designated" records , but was reduced to 13,088 unique HPSA "Designated" records for final deliverable files.

4) Create HPSA Boundary Files

All necessary geographic boundary files (Census 2010 County, 2010 Minor Civil Division, and 2010 Census Tract) were downloaded from the Census website. They were aggregated and combined so that there was only one unique record per FIPS code with no duplicates. This was necessary to preserve the record count when the Census geography was joined to the data from the HPSA definitionsfile.

The designation Excel file was split into three interim files using the **Discipline_Class_Description** for Dental Care, Mental Health, and Primary Care. *"HPSA_DC_D.TAB"* had 1,973 records marked Dental Care, *"HPSA_MH_D.TAB"* had 4,779 records marked Mental Health, and *"HPSA_PC_D.TAB"* had 6,336 records marked Primary Care.

The mappable objects in the three provider type files were made from Census polygons. A series of Structured Query Language (SQL) joins were completed to append the Census boundary polygons to the *"HPSA_XX_D.TAB"* files (where XX are PC, DC, or MH) based on the HPSA field **HPSA_Geography_ID** matching the **Geoid10** field in the Census files. The process was repeated for Census 2010 Counties, Census 2010 Minor Civil Divisions, and Census 2010 Tracts so that all HPSA Areas in the U.S. and Puerto Rico were joined with boundaries. The updated files with boundaries were renamed *"HPSA_XX_BD.TAB"* for ***B***oundary ***D***esignated.

There were a few HPSA Single County Areas in outlying U.S. Territories that needed to be analyzed with a different process because Census boundaries are not available to match to ZIP Code boundaries. (See table on page 3). Dental Care had 32 unmatched records in American Samoa (5), Federated States of Micronesia (4), Republic of Palau (16), Northern Mariana Islands (4), and U.S. Virgin Islands (3). Mental Health also had 32 unmatched records in American Samoa (5), Federated States of Micronesia (4), Republic of Palau (16), Northern Mariana Islands (4), and U.S. Virgin Islands (3). Primary Care had 63 unmatched records in American Samoa (5), Federated States of Micronesia (3), Guam (1), Marshall Islands (31), Republic of Palau (16), Northern Mariana Islands (4), and U.S. Virgin Islands (3).

The polygons were cleaned for topological errors using proprietary Check Region software. The three files were renamed *"HPSA_XX_BD_CR.TAB."* to indicate that they had been cleaned. The

HPSA boundary creation process was complete. The three 'CR' files were used as the HPSA boundaries throughout the remaining processes.

5) Prepare Base ZIP Code File

The MultiNet 5-Digit ZIP Code file *"USzips.TAB"* was opened and renamed *"USzips201406.TAB"* to clearly indicate the data vintage. Seven fields were added to the record layout to hold the results of the HPSA analysis. (USPS_Dom_SinglCo, Pct_In_Primary, Pct_or_Dom_Primary, Pct_In_Dental, Pct_or_Dom_Dental, Pct_In_Mental, and Pct_or_Dom_Mental) The *"USzips201406.TAB"* file was run through the proprietary Check Regions process, but no cleaning or changes were needed. No other additions, alterations, or validity checks were made to the standard MultiNet 5-Digit ZIP Code product.

6) Calculate Percentage Overlap for 'Region' ZIP Codes

The decision criteria for which ZIP Codes are eligible for automatic bonus payments were confirmed with our Subject Matter Expert and by CMS staff. A basic two step process was followed to determine inclusion on the lists of eligible ZIP Codes. First, a geographic test determined whether or not each ZIP Code was located completely within the respective HPSA boundary. "Completely within" was defined as equal to or greater than 99% within the respective HPSA boundary. Previously, the 99% threshold was chosen because of the non-overlapping slivers caused by geographic imprecision between boundary files created from different sources. The second test, based on USPS Dominant Counties, will be discussed later in step nine.

'Region' ZIP Codes, those ZIP Codes that have a geographical area rather than a single point, were temporarily selected out of the file *"USzips201406.TAB,"* and compared with the HPSA boundary files like *"HPSA_PC_BD_CR.TAB."* The geographic proportion of overlap was calculated where each ZIP Code intersects (or was within) each individual HPSA boundary for each of the three provider type classes. The percentage results were stored in three temporary tables, and each was summed by ZIP Code to determine how much each ZIP Code overlapped any or all HPSA boundaries it touches.

7) Find and Mark 'Point' ZIP Codes within HPSA Boundaries

'Point' ZIP Codes are 5-Digit ZIP Codes that have no defined areas, and that, as a result, appear as dots on a map without boundaries. These are represented in the 5-Digit ZIP Codes product as points rather than polygons. Examples of point ZIP Codes include PO (Post Office) Box ZIP Codes, RPOs (Residential Post Offices), and unique ZIP Codes (single sites, buildings or organizations).

The MultiNet ZIP Code data includes an **Enclosing_Zip** field specifying which 'region' ZIP Code each point zip falls within. The document describing the process used in 2010, *"READ ONLY - Sample Processes for SOW HHSM-500-2011-00015C (2-16-11).pdf "* says, "Point ZIP Codes were assigned the HPSA overlap status of their parent ZIP Code and added to the lists."

In other words, the previous contractor used the **Enclosing_Zip** tabular information to make assignments, rather than a geographic selection.

ZIP Code regions can be large areas, and can include diverse populations that may make the 'parent assignment' method inaccurate. A point on a map is by definition either fully within, or completely outside any boundary on the map. For example, when a ZIP Code region falls partially within a HPSA boundary, it is excluded from automatic payment eligibility, but if one or more 'point' ZIP Codes of that 'parent' are actually located within the HPSA boundary, they should be eligible. We were directed to use the 'most conservative' method. This project used a geographical query to select all 'point' ZIP Codes that are ***actually located*** within HPSA boundaries. The selection results were stored in three temporary tables, and each point ZIP Code within a HPSA Area was assigned a value of 100% overlap.

8) Update 'Region' and 'Point' Percentage Overlap Into Master USzips201406 Table

The summed percentage amounts from the three 'region' ZIP Code temporary tables were updated into three fields in *"USzips201406.TAB,"* **Pct_In_Primary**, **Pct_In_Dental**, and **Pct_In_Mental**. The 100% percent values from the three 'point' ZIP Code temporary tables were similarly updated into three fields in *"USzips201406.TAB,"* **Pct_In_Primary**, **Pct_In_Dental**, and **Pct_In_Mental**. The first geographic test part of the two part determination process was complete.

9) Reprocess Zips affected by duplicate HRSA boundaries using combined object method

HPSA Areas can be made from any combination of Census boundary types, and are not required to be unique. In some cases, duplicate or overlapping HPSA boundaries can cause double counting with the percentage overlap calculation method used in Step 6. Duplicate HPSA boundaries are very few in number, and can be corrected with an alternative calculation based on combining all HPSA boundaries into a single geographic object.

Similarly, there are a few documented duplicate ZIP Codes in *"USzips201406.TAB."* that could cause double counting percentage overlap calculations. There are only five non-unique ZIP Codes. They are included in the ZIP Code table twice because they cross a state boundary. Since they are actually in two states, it would be incorrect to only attribute one state FIPS code to the Zip, so TomTom elected to include each of the five ZIP Codes twice with both state FIPS Codes.

A series of quality control steps were run to verify the data and recalculate the geographic percentage overlap for any ZIP Codes that could potentially be calculated incorrectly with the method in Step 6. The following process was run for each provider type; Dental Care, Mental Health, and Primary Care.

The specific procedure is to query all of the duplicate HPSA boundaries (for each provider type) and save them as tables *"HPSA_XX_BD_CR_dups.TAB."* Then query all of the ZIP Codes that intersect the duplicate HPSA boundaries into table *"USzips201406_INT_XX_CR_dups.TAB."* These are all of the ZIP Codes that could be affected. Next, find all of the HPSA Area parts that intersect or touch the potentially affected ZIP Codes so they are completely surrounded. Combine the subset of all the geographic HPSA Area objects into a single geographic object. Save the combined HRSA subset as *"HPSA_XX_BD_CR_INTzip_subset_1obj.TAB."* Recalculate the percentage overlap for the Zip Codes using the single object file. Calculating with the single object file eliminates any potential duplicated calculation errors. Last, update the percentage recalculations into *"USzips201406_D_QC.TAB."*

Repeat similar process to find and recalculate the percentage overlap for duplicate ZIP Codes in *"USzips201406.TAB."* Update the percentage recalculations into *"USzips201406_D_QC.TAB"* for ***D***esignated - ***Q***uality ***C***ontrol.

10) Update USPS Dominant County Flags and Overwrite Geographic Overlap

The second decision criteria test to determine which ZIP Codes are eligible for automatic bonus payments is based on the County the ZIP Code is within. The 'County of Dominance' is indicated for each listed ZIP Code according to the U.S. Postal Service (USPS). A file named *"HRSA_ZIPS_OCT14.txt"* was provided by CMS that contained ZIP Code and County FIPS pairs in a single column of 10 digit numbers representing 5 digit ZIP Codes concatenated with 5 digit FIPS Codes.

The unique county FIPS Codes were identified in *"HRSA_ZIPS_OCT14.txt,"* and marked to indicate which FIPS Codes were Designated and Single County types used to make the three provider type HPSA boundary files *"HPSA_XX_BD_CR."* Temporary tables were saved with all ZIP Codes and county FIPS Codes that are used in HPSA boundary definitions *AND* listed as a ZIP Code in a USPS Dominant County. The three temporary tables were visually reviewed with the HPSA boundaries in maps, and statistically analyzed for total counts.

Fields **USPS_Dom_SinglCo, Pct_or_Dom_Dental, Pct_or_Dom_Mental** and **Pct_or_Dom_Primary,** were previously added to *"USzips201406_D_QC.TAB,"* to hold the County of Dominance results. The field **USPS_Dom_SinglCo** was updated to show "True" for all ZIP Codes that that are on the *"HRSA_ZIPS_OCT14.txt"* list *AND* used to make HPSA boundaries for any provider type.

The three 'Percent or Dominant' fields like **Pct_or_Dom_Primary** were updated in a two step process. First, the area overlap calculation was copied from **Pct_In_Primary** to **Pct_or_Dom_Primary.** This established the area geographic 99% baseline for the two part test.

All ZIP Codes were selected that match the temporary tables with all ZIP Codes and county FIPS Codes that are used in HPSA boundary definitions *AND* listed as a ZIP Code in a USPS Dominant County. The selection updated the field **Pct_or_Dom_Primary** with the numeral 2, thus overwriting any percentage that may have been in the field from step one. This two step process marks all ZIP Codes with the greater of its area overlap, or if it is in a USPS Dominant County.

Last, update field **Pct_or_Dom_Primary** for ZIP Codes that are on the USPS Dominance list *AND* have a different FIPS Code than listed on the USPS Dominance list with the numeral 3.

This usually indicates that the ZIP Code is geographically located outside the County HPSA Area border, but should be included by virtue of the County's Dominance over the ZIP Code.

11) Make Workspace Map to check for Obvious Errors

We split up the master ZIP Code file into 9 queries (3 provider type classes => Dom, 99%, Part) and visually reviewed for obvious errors. MapInfo ProViewer (version12.0) was downloaded for the non-GIS reviewer as a QC tool. The GIS specialist created four files – the master file with all zips and three subsets by provider type – that are viewable through the software. HPSA boundaries are highlighted with yellow fill. Within the yellow space, the areas with diagonal lines follow county of dominance rule and the areas without follow 99% coverage rule. Single point ZIP Codes are shown as stars. When we hover over the surface with the info tool, we can access the generated zip code records with all their fields as well as the information from the HRSA designation file. To provide for ongoing support in response to inquiries, we have appended informational fields indicating where that record is accessed and what Census source is used to map it. The information fields are **OrigRow** (Original Row); **DonlyROW** (Designated Only Row); **GIS_Boundary_Source** (GIS Boundary Source - namely Census file). The MapInfo ProViewer is free and a great tool for checking data visually through maps.
http://www.pbinsight.com/support/product-downloads/item/mapinfo-proviewer-v12.0

12) Make DBF lists of ZIP Codes from Pct_or_Dom_Primary type fields greater than 99%

The decision criteria for which ZIP Codes are eligible for automatic bonus payments is greater than 99% within the respective HPSA boundary, or listed as a ZIP Code in a USPS Dominant County used to define a Single County HPSA. ZIP Codes matching the criteria were queried out of the three fields like **Pct_or_Dom_Primary,** and saved as MapInfo DBF format files. The DBF files were opened in Excel and saved in Excel format as the three final files like *"FullyPriCareHPSA_20140829.xlsx"*.

13) Edit the Excel File lists, Add U.S. Territories Zip Codes, Remove Non-Existent Zips

ZIP Code boundaries available for this project covered all of the 50 states, the District of Columbia, Puerto Rico, Guam, and U.S. Virgin Islands. HPSA designations are also found in other outlying areas and territories of the United States, e.g.: American Samoa, Federated States of Micronesia, Republic of Palau, and Northern Mariana Islands. The qualifying geographic components of HPSA designation in these areas were whole counties/county equivalents. These areas are served by the U.S. Postal Service and do have ZIP Code boundaries available. The USPS.gov website reports ZIP Code to county correspondence for these areas and their online database is up-to-date. These relationships were evaluated for HPSA overlap, and appended to the GIS data results. Please refer to the outlying region table for details on page5.

QUALITY CONTROL

Besides checking each step of the way as we created and ran the programs, we completed the following QC steps:

- Verify HRSA data warehouse for accuracy: HPSA by state and county (http://hpsafind.hrsa.gov/) and HPSA by Address (http://datawarehouse.hrsa.gov/GeoAdvisor/ShortageDesignationAdvisor.aspx).
- Reviewed against lists of earlier years posted on the CMS website for comparisons and investigated on the differences with lists produced in previous years.
- Non-GIS QC person reviewed the results in MapInfo ProViewer software for error checking (see details in Task 11 on page12).
- Performed rigorous testing and review at each step of the task.
- Communicated with CMS/HRSA for verifications of data supplied and definitions of the key terminologies.

INPUT FILES

Input Data for Identification of ZIP Code Areas Eligible for HPSA Bonus Payments, 2014 Input Files Date of Information

Source Description
Census Boundary Files
2010
United States Census Bureau
Boundaries for Counties, Census Tracts, and Minor Civil Divisions for the U.S., Puerto Rico, Guam, and the U.S. Virgin Islands.

TomTom MultiNet (Pitney Bowes MapInfo) ZIP Code Boundary Files
June 2014
ZIP Code Boundaries for the U.S., Puerto Rico, Guam, and the U.S. Virgin Islands.

Health Professional Shortage Area (HPSA) Designation File
August 29, 2014
HRSA Data Warehouse
(http://datawarehouse.hrsa.gov), accessed September 20, 2014)
Data base of HPSA designations for all health disciplines. Inventorying the health discipline (Primary Care, Mental Health, etc), HPSA type (Single County, Service Area, etc), geographic extent, designation dates, and other characteristics.

USPS ZIP Code Dominance File
File Acquired and Supplied by CMS
October 2014
United States Postal Service
County of Dominance for ZIP Codes in the U.S. and Outlying Areas